

Irodov Problem 1.13

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1.13 Point A moves uniformly with velocity v so that the vector $\vec{v}$ is continually ‘aimed’ at point B which in its turn moves rectilinearly and uniformly with velocity $u < v$. At the initial moment of time $\vec{v} \perp \vec{u}$ and the points are separated by a distance l . How soon will the points converge?

I want to discuss the solution of the above problem from ‘Problems in General Physics’ by I.E. Irodov [1]. Let T be the time taken for points A and B to converge. So, we need to find an expression for T in terms of the quantities u , v , and l .

1 School kid’s solution

Let r be the distance between the points A and B, and θ be the angle between the velocities $\vec{v}$ and $\vec{u}$. We consider the relative velocity of A with respect to B along AB and orthogonal to AB to obtain the following relation.

$$\dot{r} = -v + u \cos \theta \quad (1)$$

$$r\dot{\theta} = -u \sin \theta \quad (2)$$

First, we will eliminate $\dot{\theta}$ by differentiating equation 1 with respect to time and then substituting the expression for $\dot{\theta}$ from equation 2.

$$\begin{aligned} \ddot{r} &= -u\dot{\theta} \sin \theta \\ \implies r\ddot{r} &= u^2 \sin^2 \theta \end{aligned}$$

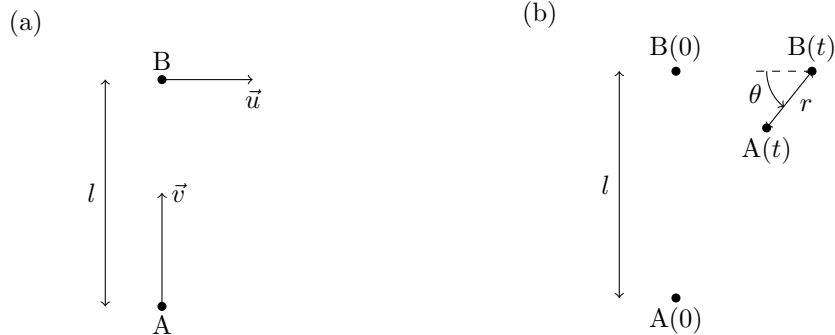


Figure 1: (a) Velocities of points A and B separated by a distance of l at the initial moment of time $t = 0$ (b) Velocities of points A and B separated by a distance of r at time $t \in (0, T)$.

Now, we will eliminate θ from the above expression by using the trigonometric identity $\cos^2 \theta + \sin^2 \theta = 1$, and equation 1.

$$\begin{aligned} r\ddot{r} &= u^2 \sin^2 \theta \\ &= u^2 (1 - \cos^2 \theta) \\ &= u^2 \left\{ 1 - \left[\frac{\dot{r} + v}{u} \right]^2 \right\} \end{aligned}$$

Simplifying the above expression we obtain the following second order differential equation.

$$\begin{aligned} r\ddot{r} + \dot{r}^2 + 2\dot{r}v + v^2 - u^2 &= 0 \\ \implies \frac{d}{dt}(r\dot{r} + 2rv) + v^2 - u^2 &= 0 \end{aligned}$$

Now, integrating with respect to time from $t = 0$ to T we get the following expression.

$$\begin{aligned} \int_{t=0}^{t=T} \frac{d}{dt}(r\dot{r} + 2rv) dt + \int_{t=0}^{t=T} (v^2 - u^2) dt &= 0 \\ \implies r(T)\dot{r}(T) + 2r(T)v - r(0)\dot{r}(0) - 2r(0)v + (v^2 - u^2)T &= 0 \end{aligned}$$

We will use the fact that

$$\begin{aligned} r(0) &= l \\ \dot{r}(0) &= -v + u \cos \pi/2 && \text{(from equation 1)} \\ &= -v \\ r(T) &= 0 && \text{(since A and B converge at time T)} \end{aligned}$$

to get the final expression for T .

$$T = \frac{lv}{v^2 - u^2} \quad (3)$$

2 Numerical solution

Interestingly, the above analytical solution does not give us the trajectory of A. Given numerical values of u , v , and l , the trajectory of point A relative to B can be found by considering the ordinary differential equations 1 and 2 to formulate an initial value problem as $r(0) = l$ and $\theta(0) = \pi/2$. On applying the forward Euler's method we arrive at the following scheme to find the trajectory of A.

$$r_{n+1} = r_n + \Delta t(u \cos \theta - v) \quad (4)$$

$$\theta_{n+1} = \theta_n - \Delta t u \sin \theta \quad (5)$$

where Δt is the time step considered for the discretisation of time, that is,

$$\begin{aligned} r_n &\approx r(n\Delta t) \\ \theta_n &\approx \theta(n\Delta t) \end{aligned}$$

Given below is a python function to simulate the motion of A relative to B.

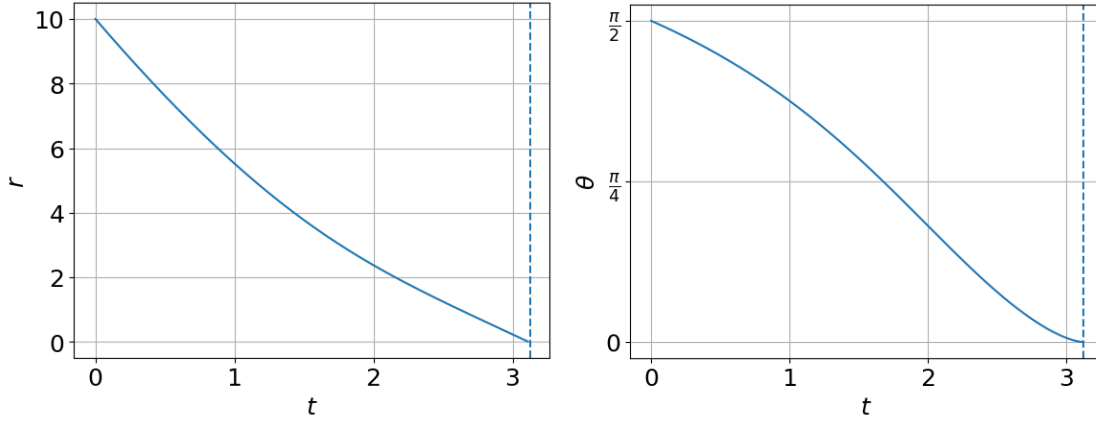


Figure 2: Simulation of r and θ for $(u, v, l) = (3, 5, 10)$, where r is the distance AB and θ the angle between the velocities of A and B. The dashed line indicates the time of contact T , according to the analytical solution given by equation 3, $T = lv/(v^2 - u^2)$.

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import numpy as np
import matplotlib.pyplot as plt

def numSol(u, v, l, dt, tolFrac = 1e-4, NMax = 100_000):
    # Initial Condition
    r = l
    th = np.pi/2

    rList = []
    thList = []

    tol = tolFrac * l
    # Forward Euler steps till r > tol
    stepCount = 0
    while r > tol and stepCount < NMax:
        stepCount += 1
        rList.append(r)
        thList.append(th)

        rNext = r + dt * (u*np.cos(th) - v)
        thNext = th - dt * u*np.sin(th) / r

        r, th = rNext, thNext

    # T_answer = stepCount * dt
    return stepCount, np.array(rList), np.array(thList)

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The consistency of the analytical result from equation 3 with the simulated relative position of A with respect to B for $(u, v, l) = (3, 5, 10)$ can be seen in figure 2. We observe this consistency for a wide range of problem parameters u , v and l .

3 Physics major's solution

Let us treat the time to collision T , as a function of r and θ . Now, given a trajectory $r(t), \theta(t)$, the rate of change of T with respect to t should be precisely -1.

$$\begin{aligned}\frac{dT}{dt} &= -1 \\ \implies \dot{r} \frac{\partial T}{\partial r} + \dot{\theta} \frac{\partial T}{\partial \theta} &= -1\end{aligned}$$

Now, using equations 1 and 2 we arrive at a partial differential equation for T .

$$\nabla T \cdot [(-v + u \cos \theta) \hat{e}_r - u \sin \theta \hat{e}_\theta] = -1 \quad (6)$$

Remember that in polar coordinates, $\nabla T = \hat{e}_r \partial T / \partial r + (1/r) \hat{e}_\theta \partial T / \partial \theta$. To pose this as boundary value problem, we will use the known values of T along the x -axis.

$$T(r, \theta) = \begin{cases} \frac{r}{v-u}, & \text{for } r > 0, \theta = 0 \\ 0, & \text{for } r = 0 \\ \frac{r}{v+u}, & \text{for } r > 0, \theta = \pi \end{cases} \quad (7)$$

Let us consider the following form of the solution for the boundary value problem described by equations 6 and 7.

$$T(r, \theta) = R(r)A(\theta)$$

From the boundary values, equation 7, it follows that $R(r)A(0) = r/(v+u)$, hence we can choose $R(r) = r$ and absorb other constant factors in $A(0)$. This linearity of R can also be justified from equation 6 as $(-v + u \cos \theta) \hat{e}_r - u \sin \theta \hat{e}_\theta$ and the right hand side are both independent of r . Now, plugging the ansatz $T = rA(\theta)$ into the partial differential equation 6, we get a first order equation for A

$$\begin{aligned}(-v + u \cos \theta)A - u \sin \theta \frac{dA}{d\theta} &= -1 \\ \implies \frac{dA}{d\theta} + \frac{v - u \cos \theta}{u \sin \theta} A &= \frac{1}{u \sin \theta} \\ \implies \frac{d}{d\theta}(\mu A) &= \frac{\mu}{u \sin \theta}\end{aligned}$$

where

$$\mu(\theta) = \exp \int \frac{v - u \cos \theta}{u \sin \theta} d\theta \quad (8)$$

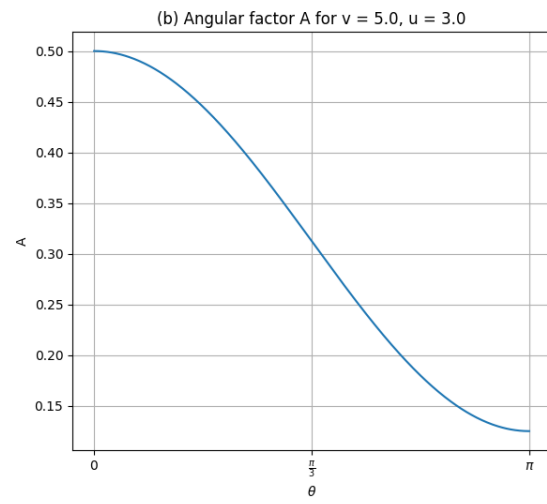
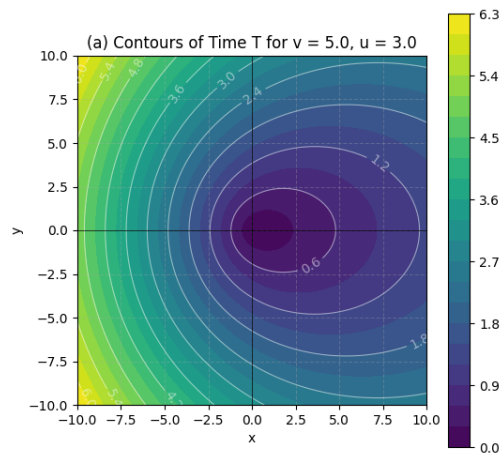
is the integrating factor. After finding μ , $A = \frac{1}{\mu} \int \frac{\mu}{u \sin \theta} d\theta$ and working out the arbitrary constants using equation 7, we get

$$A(\theta) = \frac{1}{v} \frac{1 + (u/v) \cos \theta}{1 - (u/v)^2} \quad (9)$$

$$T(r, \theta) = rA(\theta) = r \frac{v + u \cos \theta}{v^2 - u^2} \quad (10)$$

It can be verified, that this expression for T satisfies the boundary value problem defined by equations 6 and 7. From mathematical arguments of existence and uniqueness we can be assured that it is indeed the correct solution.

So, for solution to the original problem, we plug $r = l$ and $\theta = \pi/2$ to get $T = lv/(v^2 - u^2)$ which is identical to 3.



Discussion

References

- [1] I. E. Irodov, *Problems in General Physics*, Mir Publishers, 1988.